1. Let p, q, and r be statement variables. Is the following argument valid? Justify your answer.

$$\begin{array}{l} p \to q \\ \sim p \to r \\ \sim q \to r \\ \therefore q \end{array}$$

Solution: We claim the argument is invalid. Here are a couple of solutions.

Suppose that the argument is invalid: the conclusion is false and the premises are true. Then from the conclusion, q is false. Then $\sim q$ is true; from the third premise r must also be true. Since q is false, p must be false to satisfy the first premise. With p false and r true, $\sim p$ is true and there is no contradiction with the second premise. Thus, it is possible to have the conclusion false while the premises are true, so the argument is invalid.

Alternatively, suppose p and q are false and r is true. Then all premises are true but the conclusion is false. Therefore, the argument is invalid.

1 mark for saying the argument is invalid.
2 marks for correct explanations.

2. Write the negation of the following statement. Determine whether the original statement or the negation is true. Justify your answer.

   $\forall x \in \mathbb{R}$, if $2x$ is irrational, then x is irrational.

   Solution:

   Negation: $\exists x \in \mathbb{R}$ such that $2x$ is irrational and x is rational.

   The original statement is true. We use a proof by contraposition.

   Contrapositive: $\forall x \in \mathbb{R}$, if x is rational then $2x$ is rational.

   Suppose that x is rational. Then $x = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$.

   Then $2x = \frac{2a}{b}$. Since $a \in \mathbb{Z}$, $2a \in \mathbb{Z}$, and so $2x$ is rational.

   1 mark for saying the original statement is true.
   1 mark for correct negation.
   2 marks for correct explanation of why the original statement is true.

3. Use the Euclidean algorithm to compute $\gcd(662, 414)$.

Solution:

$$\begin{aligned}
662 &= 414 \cdot 1 + 248 \\
414 &= 248 \cdot 1 + 166 \\
248 &= 166 \cdot 1 + 82 \\
166 &= 82 \cdot 2 + 2 \\
82 &= 2 \cdot 41 + 0
\end{aligned}$$

Therefore, $\gcd(662, 414) = 2$.

1 mark for the correct final answer.
2 marks for correct explanations.

4. Let $\{a_n\}_{n\geq 0}$ be the sequence defined by $a_0 = 0$ and

$$a_{n+1} = a_n + 2n + 1, \quad n \geq 0.$$

(a) Write down the values of a_1, a_2, a_3, a_4.

Solution:

$a_1 = a_0 + 2 \cdot 0 + 1 = 1$
$a_2 = a_1 + 2 + 1 = 4$
$a_3 = a_2 + 2 \cdot 2 + 1 = 9$
$a_4 = a_3 + 2 \cdot 3 + 1 = 16$

This is only worth 1 mark. They get 1 mark if they get all of it right. 1/2 marks if they get some of them right (even if they get only one of them right, they get 1/2 marks).

(b) Based on your answers for Part (a), guess an explicit formula for this sequence.

Solution:

$a_n = n^2$ for all $n \geq 0$

1 mark for giving a correct explicit formula.

(c) Prove that your guess is correct.

Solution:

$a_0 = 0 = 0^2$, so the guess is correct for the base case.

Assuming that $a_n = n^2$, $a_{n+1} = n^2 + 2n + 1 = (n+1)^2$, so the recurrence relation holds with the guess.

1 mark for remembering to check the base case.
2 marks for the correct explanation of the general case. (Note they can also use induction for this step.)

5. Let $A = \{0, 1, 2\}$ and $B = \{-1, 0\}$. Write down the following sets by listing their elements and using appropriate brackets.

   1 mark for each correct answer.

   (a) $A \cup B$.
   Solution: $\{-1, 0, 1, 2\}$

   (b) $A \cap B$.
   Solution: $\{0\}$

   (c) $\mathcal{P}(B)$.
   Solution: $\big\{\emptyset, \{-1\}, \{0\}, \{-1, 0\}\big\}$

   (d) $A \times B$.
   Solution: $\big\{(0, -1), (0, 0), (1, -1), (1, 0), (2, -1), (2, 0)\big\}$

6. Consider the function $f : \mathbb{R} \to \mathbb{Z}$ defined by $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor$.

(a) Determine $f(-1), f(0), f(1), f(2)$.
Solution:
$f(-1) = \lfloor \frac{1}{2} \rfloor = 0$
$f(0) = \lfloor 0 \rfloor = 0$
$f(1) = \lfloor \frac{1}{2} \rfloor = 0$
$f(2) = \lfloor 2 \rfloor = 2$

1/2 mark for getting each of these correctly.

(b) Is f one to one? Justify your answer.
Solution: No, because $f(0) = f(1)$.

1 mark for saying no. 1 mark for correct explanation.

(c) Is f onto? Justify your answer.
Solution: Yes. For any $n \in \mathbb{Z}$, take $x = \sqrt{2n} \in \mathbb{R}$. Then $f(x) = \lfloor \frac{2n}{2} \rfloor = \lfloor n \rfloor = n$, since n is an integer. So for any $n \in \mathbb{Z}$, there exists an $x \in \mathbb{R}$ such that $f(x) = n$.

1 mark for saying yes.
2 marks for the correct explanation. Note x is not unique.

7. Let $\mathbb{N}$ denote the set of positive integers. Define a relation on $\mathbb{N} \times \mathbb{N}$ by

$$(a, b) \sim (c, d) \quad \text{if and only if} \quad a + d = b + c.$$

Prove that this is an equivalence relation.

Solution:

Reflexive: For any $a, b \in \mathbb{N}$, $a + b = b + a$, so $(a, b) \sim (a, b)$.

Symmetric: Suppose that $(a, b) \sim (c, d)$. Then $a + d = b + c$, and so $c + b = d + a$. Thus, $(c, d) \sim (a, b)$.

Transitive: Suppose that $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$.

So $a + d = b + c$ and $c + f = d + e$. Thus, $a = b + c - d$ and $f = d + e - c$, and so $a + f = b + c - d + d + e - c = b + e$. Therefore, $(a, b) \sim (e, f)$.

1 mark for reflexivity.
1 mark for symmetric property.
2 marks for transitivity.
Note: The names of these properties are not important. So long as they write their meaning correctly and explain why they hold, they get the marks.

8. Solve the equation $3x - 5 = 8$ in $(\mathbb{Z}_{11}, +, \times)$.

   (Hint: You may use the fact that $3 \times 4 = 1 \mod 11$.)

   Solution: Work in the field $\mathbb{Z}_{11}$. We have

$$3x - 5 = 8 \implies 3x = 13 = 2 \implies 3 \times 4 \times x = 2 \times 4 \implies 1 \times x = 8 \implies x = 8.$$

   1 mark for correct final answer.
   2 marks for a correct explanation.

9. Let E be the set of all even integers.

   (a) Prove that $|E| = |\mathbb{Z}|$.
   Solution: We prove this by defining a bijection between these two sets.
   Let $f : \mathbb{Z} \to E$ be defined by $f(n) = 2n$.
   Since $n \in \mathbb{Z}$, $f(n) = 2n$ is even so $f(n) \in E$.
   Suppose that $f(n) = f(m)$ for some $n, m \in \mathbb{Z}$. Then $2n = 2m$, so $n = m$. Thus, f is injective.
   For any $m \in E$, take $n = \frac{m}{2}$. Since m is even, $m = 2k$ for some $k \in \mathbb{Z}$, and so $n = k \in \mathbb{Z}$. Then $f(n) = 2\left(\frac{m}{2}\right) = m$. Thus, f is surjective.

   1 mark for recognising that equality of cardinality means a bijection.
   1 mark for a correct bijection.
   1 mark for a correct proof of injectivity.
   1 mark for a correct proof of surjectivity.
   Note: alternative solutions are possible, e.g. using the Schroder–Bernstein.

(b) Prove that E is a subgroup of $(\mathbb{Z}, +)$.
Solution:

Closure: We need to show that the sum of two even numbers is even. For any $n, m \in E$, $n = 2a$ and $m = 2b$ for some $a, b \in \mathbb{Z}$. Then $n + m = 2a + 2b = 2(a + b)$. Since $a, b \in \mathbb{Z}$, $a + b \in \mathbb{Z}$, and so $n + m \in E$.

Identity: Observe that for any $n \in E$, $0 + n = n = n + 0$. Thus, 0 is the identity, and since $0 = 2 \cdot 0$, $0 \in E$.

Inverses: If $n \in E$, then $n = 2a$ for some $a \in \mathbb{Z}$, so $-2a$ is also even. Since $2a + (-2a) = 0 = (-2a) + (2a)$, every $n \in E$ has an additive inverse.

1 mark for identity.
1 mark for closure.
2 marks for inverses.
Note: slicker solutions are possible, e.g. it is enough to note that if x and y is even then so is $x - y$.

10. What is the coefficient of x^3 in the expansion of $(2x-3)^6$? You should give your answer as a single integer, and show your working.

Solution:

The binomial theorem implies

$$(2x-3)^6 = \sum_{k=0}^{6} \binom{6}{k} (2x)^k (-3)^{6-k} = \sum_{k=0}^{6} \binom{6}{k} 2^k x^k (-3)^{6-k}.$$

The coefficient of x^3 in this expansion is

$$\binom{6}{3} (2)^3 (-3)^3 = \frac{6!}{3! \cdot 3!} (8)(-27) = -4320.$$

1 mark for having the binomial coefficient $\binom{6}{3}$.
1 mark for having the factor $(2)^3(-3)^3$.
Note $\binom{6}{3}(2)^3(-3)^3$ is OK as a final answer. I don't care about -4320.

11. (a) How many distinct arrangements are there of the 10 letters of the word KOOKABURRA?
(Note: You do not need to compute an integer answer. You may give a mathematical expression that answers the question.)

Solution: There are two Ks, two Os, two As and two Rs. Thus, the number of arrangements is

$$\frac{10!}{2! \cdot 2! \cdot 2! \cdot 2!}.$$

1 mark for having 10!. 1 mark for dividing by the right thing.

(b) Suppose that an arrangement is chosen randomly from the set of all arrangements in Part (a). What is the probability that in the chosen arrangement the two A's are not next to each other?

Solution: We first find the number of arrangements where the two As are next to each other. Put the two As together as a single 'letter'. The number of arrangements is

$$\frac{9!}{2!\cdot 2!\cdot 2!}.$$

Thus, the number of arrangements with the As **not** next to each other is

$$\frac{10!}{2!\cdot 2!\cdot 2!\cdot 2!} - \frac{9!}{2!\cdot 2!\cdot 2!} = \frac{10!}{2^4} - \frac{9!}{2^3} = \frac{10! - 2\cdot 9!}{16}.$$

The probability of choosing such an arrangement is

$$\frac{\frac{10!-2\cdot 9!}{16}}{\frac{10!}{16}} = \frac{10! - 2\cdot 9!}{10!} = \frac{8\cdot 9!}{10!} = \frac{8}{10}.$$

Therefore, the probability is 80%.

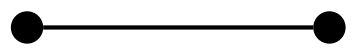
1 mark for realising that we need to do a subtraction.
1 mark for getting the number of arrangements where the two As are next to each other.
1 mark for getting the probability right.

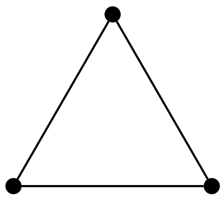
12. Let n be a positive integer. Let K_n be the graph with n vertices such that each vertex is connected to every other vertex by a unique edge (but no vertex is connected to itself, i.e. there are no loops).

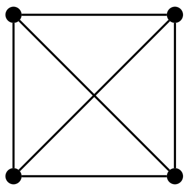
(a) Draw K_1, K_2, K_3, K_4, and K_5.

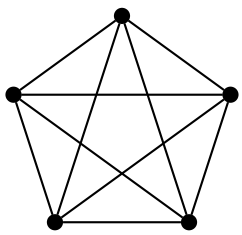
Solution:

K_1: 

K_2: 

K_3: 

K_4: 

K_5: 

1 mark for getting all of these right.
1/2 mark for getting some of them right.
(Even if they get only one right, they get 1/2 marks).

(b) Write down the adjacency matrix of K_4.

Solution:

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

1 mark for correct answer. 1/2 mark is allowed for partially correct answer.

(c) How many edges does K_n have? Justify your answer.

Solution: In the complete graph K_n, there is an edge between every pair of distinct vertices. Since there are n vertices, the number of edges (the number of ways to choose two vertices) is $\binom{n}{2}$.

Alternatively, there are n vertices, each connected to $n-1$ vertices, so there are $n(n-1)$ connections. But we counted each connection twice, so the answer is $n(n-1)/2$.

1 mark for correct final answer.
3 marks for correct explanation.

(d) For which n does K_n have an Eulerian circuit? Justify your answer.

Solution: Each of the n vertices is adjacent to each of the other $n-1$ vertices. Thus, the degree of each vertex is $n-1$. A connected graph has an Eulerian circuit if and only if every vertex has even degree. So K_n has an Eulerian circuit if and only if n is odd (to make $n-1$ even).

1 mark for remembering the condition for having an Euler circuit.
2 marks for correct explanation that this condition is satisfied.